ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 2 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 2 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Practical work is not marked strictly on syntax. Give marks for correct logic, correct pins and connections, sensible settings with reasons, and neat, labelled work. Accept other layouts and program structures that work.

QUESTION 1

1.1 IR sensor VCC, GND ✓ and OUT to a digital pin ✓; HC-SR04 VCC, GND ✓, Trig and Echo to digital pins ✓✓; servo signal to a PWM pin with 5 V and GND ✓✓; L298N ENA to a PWM pin ✓, IN1 and IN2 to digital pins ✓; motor on the L298N outputs ✓; separate 9 V motor supply to the L298N ✓ with common ground to the Arduino ✓; LED with 220 Ω to a pin ✓; neat, no short circuits ✓ (14)

1.2 Start ✓; set up pins, servo at 0°, conveyor on, LED on ✓✓; loop ✓; decision: box detected? ✓; stop conveyor, LED off ✓; measure distance, height = 30 − distance ✓✓; decision: valid reading? (error branch) ✓; decision: height > 15? ✓; Yes – diverter 90°, wait, back to 0°, tall + 1 ✓; No – short + 1 ✓; print counts ✓; wait for the box to leave, restart conveyor and loop back ✓ (14)

1.3 Circuit ✓ and flowchart ✓ saved and printed / presented (2)

[30]

QUESTION 2

2.1 Servo library and object ✓; all pin modes ✓✓; Serial.begin ✓; diverter at 0° at the start ✓; conveyor started with direction pins ✓ and analogWrite 180 ✓; LED on ✓; IR read and LOW tested ✓✓; conveyor stopped and LED off ✓✓; trigger pulse ✓; pulseIn ✓; distance calculation ✓; height = 30 − distance ✓; error check with message ✓✓; height > 15 decision ✓; servo 90°, delay, 0° ✓✓; counters increased ✓✓; counters printed ✓; wait while the box is still present ✓; conveyor restarted ✓; comments ✓ (28)

```
#include <Servo.h>
Servo diverter;
const int irPin = 2, trigPin = 7, echoPin = 8, ledPin = 12;
const int ena = 5, in1 = 6, in2 = 4;
int tallBoxes = 0, shortBoxes = 0;

void conveyor(bool on) {                  // start or stop the belt
  digitalWrite(in1, HIGH); digitalWrite(in2, LOW);
  analogWrite(ena, on ? 180 : 0);
  digitalWrite(ledPin, on ? HIGH : LOW);
}

void setup() {
  pinMode(irPin, INPUT); pinMode(trigPin, OUTPUT); pinMode(echoPin, INPUT);
  pinMode(ena, OUTPUT); pinMode(in1, OUTPUT); pinMode(in2, OUTPUT); pinMode(ledPin, OUTPUT);
  diverter.attach(9);
  diverter.write(0);                      // diverter out of the way
  Serial.begin(9600);
  conveyor(true);
}

void loop() {
  if (digitalRead(irPin) == LOW) {        // a box is at the measuring point
```

```
    conveyor(false);
    delay(300);                          // let the box settle
    digitalWrite(trigPin, LOW);  delayMicroseconds(2);
    digitalWrite(trigPin, HIGH); delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    long duration = pulseIn(echoPin, HIGH);
    int distance = duration * 0.034 / 2;  // cm
    if (distance == 0 || distance > 30) {
      Serial.println("Measurement error");
    } else {
      int height = 30 - distance;
      if (height > 15) {                 // tall box: push it into the tall lane
        diverter.write(90);
        delay(1000);
        diverter.write(0);
        tallBoxes++;
      } else {
        shortBoxes++;
      }
      Serial.print("Tall: "); Serial.print(tallBoxes);
      Serial.print("  Short: "); Serial.println(shortBoxes);
    }
    conveyor(true);
    while (digitalRead(irPin) == LOW) {}  // wait until the box has moved past
  }
}
```

2.2 Saved ✓ and printed / presented ✓ (2)

[30]

QUESTION 3

3.1 Plate 50 × 25 × 3 mm drawn ✓✓; two 16 mm holes ✓✓ spaced 26 mm centre to centre ✓ and centred on the plate ✓; flange 50 × 15 × 3 mm ✓✓ at 90° to the plate ✓; two 4 mm screw holes in the flange ✓✓; all dimensions shown ✓✓; neat drawing / clear screenshots with views ✓✓ (15)

3.2 Sensible value ✓ with a sensible reason ✓ for each setting, e.g. PLA or PETG – easy to print / strong enough; 0,2 mm layer height – good balance of strength and time; 20–40 % infill – strong enough but light; supports: no if printed flat with the flange upright, or yes for the overhang – explanation; orientation flat on the plate – best strength across the layers and no supports. (10)

3.3 Blocks: Arduino / cell counters ✓ → ESP32 ✓ (serial / I2C link named ✓) → Wi-Fi router / access point ✓ (Wi-Fi named ✓) → internet ✓ → MQTT broker ✓ (MQTT named, publish ✓) → cloud dashboard ✓ (subscribe ✓) → manager's phone or PC ✓; arrows showing the direction of the data ✓ (12)

3.4 Bracket drawing ✓, slicer table ✓ and block diagram ✓ saved and printed / presented (3)

[40]

TOTAL: 100

END OF MARKING GUIDELINE